

CROWN & ROOT

- **Crown:** The upper part of the tooth that is visible above the gumline, covered with enamel.
- **Root:** The part of the tooth that is embedded in the jawbone, covered with cementum.
- **Cementoenamel Junction (CEJ):** The boundary where the crown and root meet, also known as the cervical line.
- **Dentin:** The hard tissue that forms the bulk of the tooth beneath the enamel and cementum.
- **Pulp Chamber:** The central cavity within the crown that contains the dental pulp.
- **Pulp Canal:** The canal within the root that contains the dental pulp.
- **Pulp Cavity:** The combined space of the pulp chamber and pulp canal.

TOOTH TISSUES

- **Enamel:** The hard, outermost layer of the tooth covering the crown.
- **Cementum:** The hard tissue covering the root of the tooth.
- **Dentin:** The hard, dense tissue beneath the enamel and cementum that makes up the bulk of the tooth.
- **Pulp:** The soft tissue inside the tooth that contains blood vessels and nerves.

SUPPORTING STRUCTURES

- **Alveolar Process:** The bony ridge in the jaw that contains the tooth sockets and supports the teeth.
- **Alveolus:** The individual socket in the alveolar process that holds the tooth.
- **Gingiva:** The gum tissue that surrounds the teeth and covers the cervical third of the crown in young adults.

TOOTH SURFACES & RIDGES

- **Surfaces:** The different sides of the teeth that make contact with food or adjacent teeth.
- **Ridge:** A raised area on the surface of the tooth.
- **Incisors:** The front teeth used primarily for cutting food.
- **Canines:** The pointed teeth located next to the incisors, used for tearing food.
- **Premolars:** The teeth located between canines and molars, used for grinding food.
- **Molars:** The large, flat teeth at the back of the mouth, primarily used for grinding food.
- **Labial Surface:** The surface of incisors and canines facing the lips.
- **Buccal Surface:** The surface of premolars and molars facing the cheeks.

- **Facial Surface:** The combined labial and buccal surfaces.
- **Lingual Surface:** The surface of the tooth facing the tongue.
- **Occlusal Surface:** The chewing surface of premolars and molars.
- **Incisal Edge:** The cutting edge of incisors and canines.

PROXIMAL SURFACES

- **Mesial Surface:** The surface of the tooth that is closer to the midline of the dental arch.
- **Distal Surface:** The surface of the tooth that is farther away from the midline.
- **Contact Area:** The area where the mesial or distal surface of a tooth touches an adjacent tooth.

TOOTH TYPES

- **Anterior Teeth:** The front teeth, including central and lateral incisors and canines.
- **Posterior Teeth:** The back teeth, including premolars and molars.

ELEVATIONS ON THE SURFACE OF TEETH

- **Cusp:** An elevated point or mound on the crown of the tooth, part of the surface used for chewing.
- **Tubercle:** A small bump or elevation on the tooth surface, formed by extra enamel.
- **Cingulum:** A rounded, raised area on the lingual (tongue) surface of anterior teeth, such as incisors and canines.
- **Ridge:** A raised, linear area on the tooth surface.
 - **Marginal Ridges:** Rounded borders made of enamel found on the edges of the occlusal surface of molars and premolars.
 - **Triangular Ridge:** A slope that descends from the cusp of a molar or premolar, forming a triangular shape.
 - **Transverse Ridge:** The connection of two triangular ridges that cross the occlusal surface of the tooth.
 - **Oblique Ridge:** A ridge that crosses diagonally on the occlusal surface of maxillary molars, formed by two specific cusps (distobuccal and mesiolingual).

FOSSA

- **Fossa:** A depression or hollow area on the surface of a tooth.
 - **Lingual Fossa:** The concavity found on the lingual surface of incisors.
 - **Central Fossa:** The main depression located on the occlusal surface of molars.

- **Triangular Fossa:** Depressions located mesial and distal to the marginal ridges on the tooth.

SULCUS

- **Sulcus:** A long, narrow depression found between ridges and cusps on the tooth surface, containing a developmental groove.

LOBE

- **Lobe:** The primary section of growth that forms the crown of the tooth; cusps and mamelons are representative of lobes.

MAMELONS

- **Mamelons:** The rounded projections found on the incisal edges of newly erupted incisors, often seen in groups of three.

GROOVES

- **Developmental Groove:** A shallow line that separates the main parts of the tooth's crown or root; important for the formation of cusps and ridges.
- **Supplemental Groove:** A smaller groove that does not reach the developmental grooves; serves as an additional feature on the tooth's surface.
- **Central Groove:** The main groove that runs along the center of the occlusal surface, dividing the tooth into buccal (cheek side) and lingual (tongue side) parts.
- **Triangular Groove:** A groove that extends from the central groove towards a triangular fossa, typically found in molars.
- **Marginal Groove:** A groove located on the marginal ridges of the tooth, which facilitates the passage of food between adjacent teeth.
- **Triangular Fossa:** A depressed area located near the triangular ridges where multiple grooves converge.
- **Occlusal Pit:** A small depression where several grooves meet, usually found on the occlusal surface of molars.
- **Pits:** Tiny depressions that occur at the junctions of developmental grooves (e.g., central pit).

TOOTH ROOTS

- **Single-rooted:** Refers to teeth with one root, including all anterior teeth, the mandibular first and second premolars, and the maxillary second premolar.

- **Double-rooted:** Refers to teeth with two roots, such as the maxillary first premolar (which has buccal and lingual roots) and the mandibular molars (which have mesial and distal roots).
- **Triple-rooted:** Refers to teeth with three roots, specifically maxillary molars, which have mesiobuccal, distobuccal, and lingual roots.

DIVISION INTO THIRD

- **Crown:** The visible part of the tooth above the gum line, divided into three sections:
 - **Incisal/Occlusal third:** The top third of the crown where biting occurs (incisal for anterior teeth, occlusal for posterior teeth).
 - **Middle third:** The central portion of the crown.
 - **Cervical third:** The lower third of the crown near the gum line.
- **Root:** The part of the tooth that is embedded in the jawbone, divided into three sections:
 - **Cervical third:** The upper portion of the root, closest to the crown.
 - **Middle third:** The central portion of the root.
 - **Apical third:** The lower portion of the root, near the tip.

LINE ANGLES

- **Line Angles:** The angles formed by the junction of two surfaces of a tooth, named by combining the names of the two surfaces that meet.

LINE ANGLES OF ANTERIOR TEETH

- **Mesiolabial:** The angle formed by the mesial (closer to the midline) and labial (lip side) surfaces.
- **Distolabial:** The angle formed by the distal (away from the midline) and labial surfaces.
- **Mesiolingual:** The angle formed by the mesial and lingual (tongue side) surfaces.
- **Distolingual:** The angle formed by the distal and lingual surfaces.
- **Labioincisal:** The angle formed by the labial and incisal (cutting edge) surfaces.
- **Linguoincisal:** The angle formed by the lingual and incisal surfaces.

LINE ANGLES OF POSTERIOR TEETH

- **Mesiobuccal:** The angle formed by the mesial and buccal (cheek side) surfaces.
- **Distobuccal:** The angle formed by the distal and buccal surfaces.
- **Mesiolingual:** The angle formed by the mesial and lingual surfaces.
- **Distolingual:** The angle formed by the distal and lingual surfaces.
- **Mesio-occlusal:** The angle formed by the mesial and occlusal surfaces.
- **Disto-occlusal:** The angle formed by the distal and occlusal surfaces.

- **Bucco-occlusal:** The angle formed by the buccal and occlusal surfaces.
- **Linguo-occlusal:** The angle formed by the lingual and occlusal surfaces.

POINT ANGLES

- **Point Angles:** The angles formed by the junction of three surfaces of a tooth.

POINT ANGLES OF ANTERIOR TEETH

- **Mesiolabioincisal:** The angle formed by the mesial, labial, and incisal surfaces.
- **Distolabioincisal:** The angle formed by the distal, labial, and incisal surfaces.
- **Mesiolinguoincisal:** The angle formed by the mesial, lingual, and incisal surfaces.
- **Distolinguoincisal:** The angle formed by the distal, lingual, and incisal surfaces.

POINT ANGLES OF POSTERIOR TEETH

- **Mesiobucco-occlusal:** The angle formed by the mesial, buccal, and occlusal surfaces.
- **Distobucco-occlusal:** The angle formed by the distal, buccal, and occlusal surfaces.
- **Mesiolinguo-occlusal:** The angle formed by the mesial, lingual, and occlusal surfaces.
- **Distolinguo-occlusal:** The angle formed by the distal, lingual, and occlusal surfaces.

TOOTH DRAWING AND CARVING

- **Drawing and Carving:** The practice of creating representations of teeth, which is an integral part of studying dental anatomy.
- **G. V. Black's Table of Average Measurements:** A reference table providing standard measurements for teeth, used as a base for drawing and carving, which may be modified for improved accuracy.
- **Natural Size:** Carving teeth to their actual dimensions, using minimal fractions for precision.

PURPOSE OF CARVING AND MEASUREMENTS

- **Tooth Dimensions:** Measurements that help in comparing the sizes of different teeth.
- **Maxillary Central Incisor:** The upper front tooth, typically measuring 8.5 mm from mesial (closer to the midline) to distal (away from the midline).

- **Tooth Outlines:** The shapes of teeth recorded on squared-millimeter paper to ensure accuracy in representation.

TOOTH ASPECTS FOR COMPARISON

- **Mesial:** The surface of the tooth that is closer to the midline of the dental arch.
- **Distal:** The surface of the tooth that is farther from the midline.
- **Labial/Buccal:** The outer surfaces of the front (labial) or back (buccal) teeth facing the lips or cheeks.
- **Lingual:** The inner surface of the tooth facing the tongue.
- **Incisal/Occlusal:** The top edge of anterior teeth (incisal) or the chewing surface of posterior teeth (occlusal).

SKILLS TO DEVELOP

- **Measurement Table:** Memorizing a reference table that lists the dimensions of teeth for quick mental comparisons.
- **Accurate Outlines:** The ability to draw the correct shapes and outlines of various tooth surfaces.
- **Precision Carving:** The skill of creating tooth carvings that match the accuracy of line drawings.

MEASURING TEETH

- **Boley Gauge:** A measuring instrument used in dentistry for precise measurements of tooth dimensions.
- **Calibrations:** Specific marks on a measuring tool used to gauge the measurements of teeth accurately.
- **Key Landmarks:** Significant points on a tooth used for measurement purposes.

TEETH CATEGORIES

- **Maxillary Teeth:** Teeth located in the upper jaw (maxilla).
- **Mandibular Teeth:** Teeth located in the lower jaw (mandible).

MEASUREMENT PARAMETERS

- **Length of Crown:** The measurement of the visible part of the tooth above the gum line.
- **Length of Root:** The measurement of the part of the tooth embedded in the jawbone.

- **Mesiodistal Diameter of Crown:** The width of the crown measured from the mesial to distal surfaces.
- **Mesiodistal Diameter at Cervix:** The width of the tooth measured at the cervix (junction between the crown and root).
- **Labiolingual/Buccolingual Diameter of Crown:** The width of the crown measured from the labial/buccal surface to the lingual surface.
- **Labiolingual/Buccolingual Diameter at Cervix:** The width of the tooth measured at the cervix from the labial/buccal to the lingual surface.
- **Curvature of Cervical Line:** The shape of the gum line that curves around the neck of the tooth, measured on mesial and distal sides.

INDIVIDUAL TEETH SPECIFICATIONS

- **Central Incisor:** The front tooth, typically the largest and most prominent in the dental arch.
- **Lateral Incisor:** The tooth adjacent to the central incisor, slightly smaller.
- **Canine:** The pointed tooth located next to the lateral incisors, used for tearing food.
- **First Premolar:** The tooth located behind the canine, used for crushing and grinding food.
- **Second Premolar:** The tooth next to the first premolar, similar in function.
- **First Molar:** The first large grinding tooth located at the back of the mouth.
- **Second Molar:** The second large grinding tooth, located behind the first molar.